$$y = m_1 x_1 + m_2 x_2 + b$$

initial $m^{(0)} = [-1, 2]$

$b^{(0)} = [1, 1]$

and $x = ((1, 3), (4, 10))$

$y = (5, 6)$

$n = 2$

$k = 3$ (3 members for 3 updates)

use $\alpha = 0.01$

$$\hat{y}_1 = m_1 x_{11} + m_2 x_{12} + b_1$$
$$\hat{y}_2 = m_1 x_{21} + m_2 x_{22} + b_2$$

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \end{bmatrix} = \begin{bmatrix} x_{11} m_1 + x_{12} m_2 + b_1 \\ x_{21} m_1 + x_{22} m_2 + b_2 \end{bmatrix}$$

$$= \begin{bmatrix} x_{11} m_1 + x_{12} m_2 \\ x_{21} m_1 + x_{22} m_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$= \begin{bmatrix} x_{11} + x_{12} \\ x_{21} \quad x_{22} \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$= Xm + b$$

$\therefore \hat{y} = Xm + b$

MSE

$$J(m,b) = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Since $n = 2$

$$J(m,b) = \frac{1}{2} \left[(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 \right]$$

$e_1 = \hat{y}_1 - y_1$ and $e_2 = \hat{y}_2 - y_2$

$$e_i^2 = (y_i - \hat{y}_i)^2 = (\hat{y}_i - y_i)^2$$

$$J(m,b) = \frac{1}{2} (e_1^2 + e_2^2)$$

Recall $\hat{y} = Xm + b$ and $e = \hat{y} - y$

$\therefore e = (Xm + b) - y = Xm + b - y$

$$e = \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} \quad \text{and} \quad e^T = [e_1 \ e_2]$$

$$e^T \cdot e = \begin{bmatrix} e_1 & e_2 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = e_1^2 + e_2^2$$

Recall $J(m, b) = \frac{1}{2}(e_1^2 + e_2^2)$

$$J(m, b) = \frac{1}{2} e^T e$$

generally $$J(m, b) = \frac{1}{n} e^T e$$

since $e = Xm + b - y$

$$J(m, b) = \frac{1}{n} (Xm + b - y)^T (Xm + b - y)$$

Gradient Derivation

$$J(m, b) = \frac{1}{n}(e_1^2 + e_2^2)$$

$$e_1 = m_1 x_{11} + m_2 x_{12} + b_1 - y_1$$
$$e_2 = m_1 x_{21} + m_2 x_{22} + b_2 - y_2$$

$$\frac{\partial J}{\partial e_1} = \frac{\partial}{\partial e_1}\left[\frac{1}{n}(e_1^2 + e_2^2)\right]$$

$$= \frac{1}{n}(2e_1 + 0) = \frac{2e_1}{n}$$

$$\frac{\partial J}{\partial e_2} = \frac{\partial}{\partial e_2}\left[\frac{1}{n}(e_1^2 + e_2^2)\right]$$

$$= \frac{1}{n}[(0 + 2e_2)] = \frac{2e_2}{n}$$

So $\frac{\partial J}{\partial e_1} = \frac{2e_1}{n}$ & $\frac{\partial J}{\partial e_2} = \frac{2e_2}{n}$

$$\frac{\partial e_1}{\partial m_1} = \frac{\partial}{\partial m_1}(m_1 x_{11} + m_2 x_{12} + b_1 - y_1)$$

$$= x_{11} + 0 + 0 - 0 = x_{11}$$

$$\frac{\partial e_2}{\partial m_1} = \frac{\partial}{\partial m_1}(m_1 x_{21} + m_2 x_{22} + b_2 - y_2)$$

$$= x_{21} + 0 + 0 - 0 = x_{21}$$

So $\frac{\partial e_1}{\partial m_1} = x_{11}$ & $\frac{\partial e_2}{\partial m_1} = x_{21}$

Applying the chain rule

$$\frac{\partial J}{\partial m_1} = \frac{\partial J}{\partial e_1}\frac{\partial e_1}{\partial m_1} + \frac{\partial J}{\partial e_2}\frac{\partial e_2}{\partial m_1}$$

$$= \frac{2e_1}{n}(x_{11}) + \frac{2e_2}{n}(x_{21})$$

$$= \frac{2}{n}(x_{11}e_1 + x_{21}e_2)$$

Now

$$\frac{\partial e_1}{\partial m_2} = \frac{\partial}{\partial m_2}(m_1 x_{11} + m_2 x_{12} + b_1 - y_1)$$

$$= 0 + x_{12} + 0 - 0 = x_{12}$$

$$\frac{\partial e_2}{\partial m_2} = \frac{\partial}{\partial m_2}(m_1 x_{21} + m_2 x_{22} + b_2 - y_2)$$

$$= 0 + x_{22} + 0 - 0 = x_{22}$$

So $\frac{\partial e_1}{\partial m_2} = x_{12}$ & $\frac{\partial e_2}{\partial m_2} = x_{22}$

Applying the chain rule

$$\frac{\partial J}{\partial m_2} = \frac{\partial J}{\partial e_1}\frac{\partial e_1}{\partial m_2} + \frac{\partial J}{\partial e_2}\frac{\partial e_2}{\partial m_2}$$

$$= \frac{2e_1}{n}(x_{12}) + \frac{2e_2}{n}(x_{22})$$

$$= \frac{2}{n}(x_{12}e_1 + x_{22}e_2)$$

So $\frac{\partial J}{\partial m_1} = \frac{2}{n}(x_{11}e_1 + x_{21}e_2)$

and

$\frac{\partial J}{\partial m_2} = \frac{2}{n}(x_{12}e_1 + x_{22}e_2)$

Combining both derivaties

$$\nabla_m J = \begin{bmatrix} \frac{\partial J}{\partial m_1} \\ \frac{\partial J}{\partial m_2} \end{bmatrix}$$

$$\nabla_m J = \frac{2}{n}\begin{bmatrix} x_{11}e_1 + x_{21}e_2 \\ x_{12}e_1 + x_{22}e_2 \end{bmatrix}$$

$$= \frac{2}{n}\begin{bmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \end{bmatrix}\begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

Now, $X^T = \begin{bmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \end{bmatrix}$

So $\nabla_m J = \frac{2}{n} X^T e$

To get $\frac{\partial J}{\partial b_1}$ and $\frac{\partial J}{\partial b_2}$

Recall

$$J(m,b) = \frac{1}{n}(e_1^2 + e_2^2)$$

and $e_1 = m_1 x_{11} + m_2 x_{12} + b_1 - y_1$

$e_2 = m_1 x_{21} + m_2 x_{22} + b_2 - y_2$

$$\frac{\partial e_1}{\partial b_1} = \frac{\partial}{\partial b_1}(m_1 x_{11} + m_2 x_{12} + b_1 - y_1)$$

$$= 0 + 0 + 1 - 0 = 1$$

$$\frac{\partial e_2}{\partial b_1} = \frac{\partial}{\partial b_1}(m_1 x_{21} + m_2 x_{22} + b_2 - y_2)$$

$$= 0 + 0 + 0 - 0 = 0$$

So $\frac{\partial e_1}{\partial b_1} = 1$ & $\frac{\partial e_2}{\partial b_1} = 0$

Applying the chain rule

$$\frac{\partial J}{\partial b_1} = \frac{\partial J}{\partial e_1}\frac{\partial e_1}{\partial b_1} + \frac{\partial J}{\partial e_2}\frac{\partial e_2}{\partial b_1}$$

$$= \frac{2e_1}{n}(1) + \frac{2e_2}{n}(0)$$

$$= \frac{2e_1}{n} + 0 = \frac{2e_1}{n}$$

Now

$$\frac{\partial e_1}{\partial b_2} = \frac{\partial}{\partial b_2}(m_1 x_{11} + m_2 x_{12} + b_1 - y_1)$$

$$= 0 + 0 + 0 - 0 = 0$$

$$\frac{\partial e_2}{\partial b_2} = \frac{\partial}{\partial b_2}(m_1 x_{21} + m_2 x_{22} + b_2 - y_2)$$

$$= 0 + 0 + 1 - 0 = 1$$

So $\frac{\partial e_1}{\partial b_2} = 0$ & $\frac{\partial e_2}{\partial b_2} = 1$

then $\frac{\partial J}{\partial b_2} = \frac{\partial J}{\partial e_1}\frac{\partial e_1}{\partial b_2} + \frac{\partial J}{\partial e_2}\frac{\partial e_2}{\partial b_2}$

$= \frac{2e_1}{n}(0) + \frac{2e_2}{n}(1) = \frac{2e_2}{n}$

$\therefore \frac{\partial J}{\partial b_1} = \frac{2e_1}{n}$ & $\frac{\partial J}{\partial b_2} = \frac{2e_2}{n}$

Combining the two

$$\nabla_b J = \begin{bmatrix} \frac{\partial J}{\partial b_1} \\ \frac{\partial J}{\partial b_2} \end{bmatrix} = \begin{bmatrix} \frac{2e_1}{n} \\ \frac{2e_2}{n} \end{bmatrix}$$

$$= \frac{2}{n}\begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \frac{2}{n}e$$

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For the equation

$$m^{(k+1)} = m^{(k)} - \alpha \nabla_m J$$

$$m^{(k+1)} = m^{(k)} - \alpha \left(\frac{2}{n} X^T e^{(k)} \right)$$

and $$b^{(k+1)} = b^{(k)} - \alpha \nabla_b J$$

$$b^{(k+1)} = b^{(k)} - \alpha \left(\frac{2}{n} e^{(k)} \right)$$

recall $n = 2$, $\alpha = 0.01$

$$\alpha \frac{2}{n} = 0.01 \times \frac{2}{2} = 0.01$$

$$\therefore m^{(k+1)} = m^{(k)} - 0.01 X^T e^{(k)}$$

$$b^{(k+1)} = b^{(k)} - 0.01 e^{(k)}$$

For initial calculations

$$m^{(0)} = [-1, 2]$$
$$b^{(0)} = [1, 1]$$

$$\hat{y}(0) = X m^{(0)} + b^{(0)}$$

$$= \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1(-1) + 3(2) \\ 4(-1) + 10(2) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 + 6 \\ -4 + 20 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 + 1 \\ 16 + 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

$$\hat{y}^{(0)} = [6, 17]$$

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For residuals

$$e^{(0)} = \hat{y}^{(0)} - y$$

$$= [6, 17] - [5, 6]$$

$$= [(6-5), (17-6)]$$

$$= [1, 11]$$

For MSE

$$J_0 = \frac{1}{n}(e^{(0)})^T e^{(0)}$$

$$= \frac{1}{2}\begin{bmatrix} 1 & 11 \end{bmatrix}\begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$= \frac{1}{2}[1(1) + 11(11)] = \frac{1}{2}[1 + 121]$$

$$= \frac{1}{2}(122)$$

$$= 61$$

For Update 1

$$\nabla_m J = \frac{2}{n} X^T e^{(0)}$$

$$= \frac{2}{2} X^T e^{(0)} = X^T e^{(0)}$$

$$X^T = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \text{ or } X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

recall $e^{(0)} = [1, 11]$

$$\nabla_m J = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$= \begin{bmatrix} 1(1) + 4(11) \\ 3(1) + 10(11) \end{bmatrix} = [45, 113]$$

$$\nabla_m J = [45, 113]$$

$$\nabla_b J = \frac{2}{n} e^{(0)} = \frac{2}{2} [1, 11]$$

$$= [1, 11]$$

$$\nabla_b J = [1, 11]$$

Updating m

$$m^{(1)} = m^{(0)} - \alpha \nabla_m J$$

$$m^{(1)} = [-1, 2] - 0.01\,[45, 113]$$

$$= [-1, 2] - [0.45, 1.13]$$

$$= [(-1 - 0.45), (2 - 1.13)]$$

$$= [-1.45, 0.87]$$

$$m^{(1)} = [-1.45, 0.87]$$

$$b^{(1)} = b^{(0)} - \alpha \nabla_b J$$

$$b^{(1)} = [1, 1] - 0.01\,[1, 11]$$

$$= [1, 1] - [0.01, 0.11]$$

$$= [(1 - 0.01), (1 - 0.11)]$$

$$= [0.99, 0.89]$$

$$b^{(1)} = [0.99, 0.89]$$

$m^{(1)} = [-1.45, 0.87)$ & $b^{(1)} = [0.99, 0.89]$

$$\hat{y}^{(1)} = X m^{(1)} + b^{(1)}$$

$$= \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

$$= \begin{bmatrix} 1(-1.45) + 3(0.87) \\ 4(-1.45) + 10(0.87) \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

$$= \begin{bmatrix} -1.45 & +2.61 \\ -5.80 & +8.70 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

$$= \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix}$$

$$\hat{y}(1) = [2.15, 3.79]$$

For residuals

$$e^{(1)} = \hat{y}(1) - y$$

$$e^{(1)} = [2.15, 3.79] - [5, 6]$$

$e^{(1)} = [(2.15 - 5), (3.79 - 6)] = [-2.85, -2.21]$

MSE would then be,

$$J_1 = \frac{1}{n} (e^{(1)})^T e^{(1)}$$

$$= \frac{1}{2} \begin{bmatrix} -2.85, -2.21 \end{bmatrix} \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

$$= \frac{1}{2} [(-2.85 \times -2.85) + (-2.21 \times -2.21)]$$

$$= \frac{1}{2} [13.0066] = 6.5033$$

$m^{(1)} = [-1.45, 0.87]$

$b^{(1)} = [0.99, 0.89]$

$J_1 = 6.5033$

SECTION B : Update 2.

The starting parameters are:

$m^{(1)} = [-1.45, 0.87]$

$b^{(1)} = [0.99, 0.89]$

The residuals are:

$e^{(1)} = [-2.85, -2.21]$

The previous MSE is:

$J_1 = 6.5033$

The learning rate remains:

$\alpha : 0.01.$

Step 1: Calculate gradient respect to m

formula: $\nabla_m J = \frac{2}{n} X^T e^{(1)}$

Since $n = 2. = \frac{2}{2} X^T e^{(1)}$

$= 1 X^T e^{(1)}$

Data matrix $X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$

Substitute the previous residual vector:

$\nabla_m J = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$

Multiply the first row by residual order

$1(-2.85) + 4(-2.21)$

$= -2.85 - 8.84$

$= -11.69.$

Multiply second flow.

$3(-2.85) + 10(-2.21)$

$= -8.55 - 20.10$

$= -30.65$

combining the results.

$\nabla_m J = [-11.69, -30.65]$

Step 2: calculate gradient respect to b

The bias gradient is

$\nabla_b J = \frac{2}{n} e^{(1)}$

Substitute $n = 2. = \frac{2}{2} e^{(1)}$

$= \frac{2}{2} [-2.85, -2.21]$

$= 1 [-2.85, -2.21]$

$\nabla_b J = [-2.85, -2.21]$

Step 3: Update m.

The update equation is:

$m^{(2)} = m^{(1)} - \alpha \nabla_m J$

Substitute the values:

$= [-1.45, 0.87] - 0.01 [-11.69, -30.65]$

multiply each gradient component by 0.01

$0.01(-11.69) = -0.1169.$

$0.01(-30.65) = 0.3065$

Therefore:

$m^2 = [-1.45, 0.87] - [-0.1169, -0.3065]$

Substract corresponding components:

$= [-1.45 - (-0.1169), 0.87 - (-0.3065)]$

$= [-1.45 + 0.1169, 0.87 + 0.3065]$

$m^{(2)} = [-1.3331, 1.1765]$

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Step 4: Update b.

The bias update equation is

$b^{(2)} = b^{(1)} - \alpha \nabla_b J$

Substitute: $[0.99, 0.87] - 0.01[-2.85, -2.21]$

Multiply by the learning rate:

$0.01(-2.85) = -0.0285$

$0.01(-2.21) = -0.0221$

Therefore:

$b^{(2)} = [0.99, 0.89] - [-0.0285, -0.0221]$

$= [0.99 - (-0.0285), 0.89 - (-0.0221)]$

$= [0.99 + 0.0285, 0.89 + 0.0221]$

$b^{(2)} = [1.0185, 0.9121]$

The prediction equation is

$\hat{y}^{(2)} = Xm^{(0)} + b^{(2)}$

Substitute

$\begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} + \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$

Perform matrix multiplication:

$= \begin{bmatrix} 1(-1.3331) + 3(1.1765) \\ 4(-1.3331) + 10(1.1765) \end{bmatrix} + \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$

Calculate the first component:

$1(-1.3331) = -1.3331$

$3(1.1765) = 3.5295$

$-1.3331 + 3.5295 = 2.1964$

Calculate the second component:

$4(-1.3331) = -5.3324$

$10(1.1765) = 11.7650$

$-5.3324 + 11.7650 = 6.4326$

So, $\hat{y}^{(2)} = \begin{bmatrix} 2.1964 \\ 6.4326 \end{bmatrix} + \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$

Add corresponding components:

$= [2.1964 + 1.0185, 6.4326 + 0.9121]$

$\hat{y}^{(2)} = [3.2149, 7.3447]$

Step 6: Calculate residuals after Update 2.

The residual equation is

$e^{(2)} = \hat{y}^{(2)} - y$

Substitute:

$= [3.2149, 7.3447] - [5, 6]$

Subtract:

$= [3.2149 - 5, 7.3447 - 6]$

$e^{(2)} = [-1.7851, 1.3447]$

Step 7: Calculate MSE after update 2

The cost function is:

$J_2 = \frac{1}{n}(e^{(2)})^T e^{(2)}$

Since $n = 2$:

$\frac{1}{2}\begin{bmatrix} -1.7851 & 1.3447 \end{bmatrix} \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$

Multiply row by column:

$= \frac{1}{2}[(-1.7851)(-1.7851) + (1.3447)(1.3447)]$

Calculate the squared residuals:

$(-1.7851)^2 = 3.18658201$

$(1.3447)^2 = 1.80821809$

Add: $3.18658201 + 1.80821809$

$= 4.99480010$

Divide by the number of observations:

$J_2 = \frac{4.99480010}{2}$

$J_2 = 2.49740005$

Result after Update 2:

$m^{(2)} = [-1.3331, 1.1765]$

$b^{(2)} = [1.0125, 0.9121]$

$J_2 = 2.49740005$

The MSE decreased from 6.5033 to 2.49740005 showing that the second update further improved the overall predictions.

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SECTION C MERVEILLE INECA

our starting parameters are: $m^{(2)} = [-1.3331, 1.1765]$

$b^{(2)} = [1.0185, 0.9121]$

The residuals are: $e^{(2)} = [-1.7851, 1.3447]$

The previous MSE is: $J_2 = 2.49940005$

The learning rate remains: $\alpha = 0.01$

Formula: $\Delta_m J = \frac{2}{n} X^T E$, $n = 2 = \frac{2}{2} X^T e = 1 X e^T$

$\Delta_m J = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix} = 1[-1.7851] + 4[1.3447] = -1.7851 + 5.37$

$= 3.5937$

$3[-1.7854] + 10[1.3447] = -5.3553 + 13.4470 = 8.0917$

$\Delta_m J = [3.5937, 8.0917]$

Step 2: $\Delta_b J = \frac{2}{n} e = \frac{2}{2}[-1.7854, 1.3447] = 1[-1.7851, 1.3447$

$= [-1.7854, 1.3447]$

Step 3: Update m $m^{(3)} = m^{(2)} - \alpha \Delta_m J = [1.3331, 1.1765] - 0.01[3.5937 8.0$

$0.01[3.5937] = 0.035937$, $0.01[8.0917] = 0.080917$

$m^{(3)} = [-1.3331, 1.1765] - [0.035937, 0.080917]$

$= [-1.3331 - 0.035937, 1.1765 - 0.080917]$

$m^{(3)} = [-1.369037, 1.095583]$

Step 4: update b $b^3 = b^{(2)} - \alpha \Delta_b J$

$= [1.0185, 0.9121] - 0.01[-1.7851, 1.3447]$

$0.01[-1.7851] = -0.017851$, $0.01(1.3447) = 0.013447$

$b^{(3)} = [1.0185, 0.9121] - [-0.017851, 0.013447]$

$b^{(3)} = [1.036351, 0.089853]$

$m^{(3)} = [-1.369037,\ 1.095583]$

$b^{(3)} = [1.036351,\ 0.898653]$

$^1y^{(3)} = Xm^{(3)} + b^{(3)}$

$= \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.369037 \\ 1.095583 \end{bmatrix} + \begin{bmatrix} 1.036351 \\ 0.898653 \end{bmatrix}$

$= \begin{bmatrix} 1(-1.369037) + 3(1.095583) \\ 4(-1.369037) + 10(1.095583) \end{bmatrix} + \begin{bmatrix} 1.036351 \\ 0.898653 \end{bmatrix}$

$= 1(-1.369037) = -1.369037$

$= 3(1.095583) = 3.286749 = -1.369037 + 3.286749 =$

1.917712

$4(-1.369037) = -5.476148 = 10(1.095583) = 10.955830$

$= -5.476148 + 10.955830 = 5.479682$

$^1y^{(3)} = \begin{bmatrix} 1.917712 \\ 5.479682 \end{bmatrix} + \begin{bmatrix} 1.036351 \\ 0.898653 \end{bmatrix}$

$= [1.917712 + 1.036351,\ 5.479682 + 0.898653]$

$^1y^{(3)} = [2.954063,\ 6.378335]$

Step 6: $e^{(3)} = {}^1y^{(3)} - y$

$= [2.954063 - 5,\ 6.378335 - 6]$ $e^{(3)} = [-2.045937,\ 0.378335]$

Step 7: $J_3 = \frac{1}{n}(e^{(3)})^T e^{(3)}$, $n = 2$

$= \frac{1}{2} \begin{bmatrix} -2.045937 & 0.378335 \end{bmatrix} \begin{bmatrix} -2.045937 \\ 0.378335 \end{bmatrix}$

$= \frac{1}{2} \begin{bmatrix} (-2.045937)(-2.045937) + (0.378335)(0.378335) \end{bmatrix}$

$(-2.045937)^2 = 4.185858207969$, $(0.378335)^2 = 0.143137372225$

$4.185858207969 + 0.143137372225 = 4.328995580194$

$J_3 = \frac{4.328995580194}{2} = 2.164497790097$

Result: $m^{(3)} = [-1.369037,\ 1.095583]$

$b^{(3)} = [1.036351,\ 0.898653]$

$J_3 = 2.164497790097$

The MSE decreased from 61 initially to approximately 6.5033, 2.4974 and 2.1645 after the three updates.

Although some parameters changed direction between updates, the error decreased each time.